

Problem Statement ver. 8/10/23

As a team, complete the assignment outlined below. Include a cover sheet with the assignment submittal and include a brief executive summary. The cover sheet for any team-based senior design submittal should include the name of the submittal, the name of the project, the date, and the name(s) of all participants.

Part A: Problem Statement

First, define the problem. The problem statement is one sentence (ideally) that clearly and concisely defines the problem to be solved. It is stated in the language of the customer or end user of the project item. It might be necessary to provide additional paragraphs that provide context to the problem. This helps define the scope and/or the customer or end-user.

Part B: Requirements and Creating Value

Describe requirements the customer or end-user would expect of the design project. These should be high-level requirements that the design project should satisfy. Describe how this design will create value for the customer or end-user.

Part C: Existing Solutions

Next, include a section on existing design solutions you have found. What part(s) of your problem do they solve? What aspects or parts of the problem are not satisfied? What are the requirements and specifications of these existing solutions, if available? What are their costs? What are their shortcomings? Do they comply with any engineering standards (if so, identify the standards with which they comply)?

Carefully consider how you organize this information. How should you best communicate your findings? What format(s) should you use? Tables are great ways to provide information for comparison.

Part D: Solution

Finally, describe your team's initial thoughts and possible approach to providing a solution to the problem. Identify the benefits offered by your intended solution, how your solution will create value, and how it differs from any existing solutions.

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Problem Statement Grading

Category	Below	Adequate	Above	Comments
Real problem identified	☐	☐	☐	_____
Clearly defines problem	☐	☐	☐	_____
Concisely defines problem	☐	☐	☐	_____
Stated in the customer's language	☐	☐	☐	_____
Design project requirements identified	☐	☐	☐	_____
Envisioned solution and its benefits	☐	☐	☐	_____
Existing solutions: How solved	☐	☐	☐	_____
Existing solutions: Requirements/Specs	☐	☐	☐	_____
Proposed Solution: Benefits	☐	☐	☐	_____
Proposed Solution: Value to customer	☐	☐	☐	_____
Writing quality	☐	☐	☐	_____

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Problem Statement

Smart Home Manager

09/13/26

Miles Bailey-Waltz

Jackson Duty

Megan Ruffing

Lily Schellenbach

Atifa Tariq

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Part A: Problem Statement

Energy is often wasted due to fans and lights left on in a room when no one is there. Additionally, setting the temperature manually often causes the room to be either too hot or too cold. Our room manager can track each of these variables and automatically set them to maintain a comfortable room environment while also saving power.

Part B: Requirements and Creating Value

The customer target of this product would be ~~homeowners or building managers, or even architect/construction companies~~ current homeowners, building managers, or any companies constructing new residential units. Customers would expect the system to accurately monitor room conditions and occupancy while automatically adjusting the temperature and lighting. The design should be reliable, energy efficient, easy to use, and be capable of notifying users when unusual spikes in energy consumption or unsafe room conditions are detected. ~~This~~ Our design would create value by reducing unnecessary energy use, maximizing occupant comfort and convenience, and most importantly, lowering the customer's utility costs.

Part C: Existing Solutions

Most existing temperature and environment monitoring systems are built for non-residential applications to prevent power outages or control temperature. There is a small market for room management systems, but most exclusively control temperature, or track the number of people in a room. There is no single product on the market that can monitor temperature, automate light and fan control, and track occupants.

Room Alert has created a product for server rooms that can monitor temperature, humidity, and power. This product is tuned for server rooms rather than a residential home. It is not capable of automatic light and fan control, nor can it track the number of people in a room. This device is also quite costly, with a central hub that costs a minimum of \$779 for a maximum of 3 sensors, and each temperature sensor costing an additional \$79. Room alert also uses a subscription model for their web dashboard to control these sensors, costing even more for continued use (<https://account.roomalert.com/pricing>).

For a residential temperature monitoring system, Ecobee is a common choice. The thermostat costs \$140 and controls the temperature in multiple rooms. This does not include occupancy sensing, fan control, or lighting control.

Part D: Solution

Our first thoughts include a main system that controls the temperature and lighting based off external sensors that we will need to install either by routing cable or Bluetooth. The benefits offered by this are the reduction of electric and gas bills. It will also seamlessly increase comfort for users by adjusting the air conditioning, fans, and lights depending on the time of day, occupancy, and current temperature. Our solution would create value by combining these

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multiple technologies that aren't typically one system. There are solutions for server rooms but a comprehensive system like this has not been fully brought into residential applications. Our project is unique because there are some systems for either a lighting system or a heating and cooling system but there are not any that have both for the home.